Fruit Fly

Team: Cheesecake

Element D: Design Concept Generation, Analysis, and Selection

Our boarding school residents need a way to remove fruit flies from their dormitories. This is because residents are irritated by the fruit flies and they could also pose a health risk. The fruit flies are attracted to a food item that was left in the dormitories a few weeks ago. Fruit flies have a short reproduction time so the quicker they are removed the easier it will be because there are less flies to remove. Quality of life in the dormitories would also improve if the fruit flies were to be removed.

What we want to do is make a contraption that will smell fermented for the flys. They will go into the hole and not be able to get out. We made our design a cup shape and added a screw that will screw onto the bottle. There will be a funnel that will allow the flys to fly in and get stuck.

We figured out a way to trap them by using other popular traps and reversing the initial intention. We all liked the idea and started to sketch out what it would look like. We ended up adding a lot of new features in the end which made our design even better. For the final solution, we added a funnel and screw cap for the fly trap.

Our final design was 3D printed and turned out really great. The size of the print was the right size and the bottle cap screws on tight. Our final solution was chosen because we wanted our fly trap to be completely different from other fruit traps. Being able to print a prototype before our final one helped give more ideas as well as what needed to be improved. The design requirements were to get rid of 80% of the flies. We needed to also make sure the fly trap would not break if it was touched or dropped.

The brainstorming is where we started and we decided to make the trap have a funnel for the flys to fly into. The fly trap also needs a low weight so we 3D printed it. We chose the 3D printer because it would be quick but also detailed. For our design, the 3D printer creates a prototype that is within the weight requirements.

The sketches had new items that we wanted to add. These are a screw on cap and also a built in funnel. We wanted to build it out of cardboard at first but with the other materials we were adding we did not want the cardboard to get wet. We wanted to add changes to the design and add onto the funnel but this was at the last second. We wanted the funnel to be able to screw on and be easy to take off. We plan on doing that as another prototype to test.